

Digital Egypt Pioneers Initiative

Round 4  
2025 - 2026

رواد مصر الرقمية

الدورة الرابعة  
2025 - 2026

**Track:** AI & Data Science - Microsoft Machine Learning Engineer

**Group Code:** ONL4\_AIS2\_S7

**Instructor:** Eman Elgalad

**Company Name:** Skills Dynamix

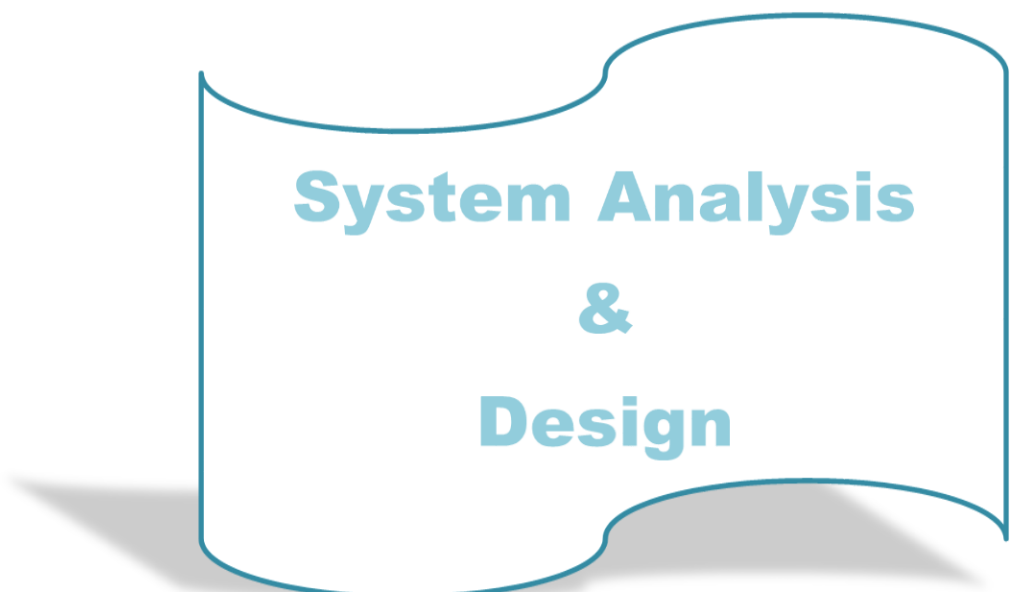


**Project Name**

**"AI-Powered Predictive Maintenance for Industrial Equipment"**

| No. | Name                               |
|-----|------------------------------------|
| 1   | Moaz Osama Esamil Hataba (TL)      |
| 2   | Abrar Ashraf Shaban Ahmad          |
| 3   | Marwan Muhammad Darwish Ammar      |
| 4   | Yussef Ahmad Muhammad Elshamy      |
| 5   | Salma Muhammad Nageh Abdul-Hamid   |
| 6   | Nour-Eldin Kamel Muhammad Hasanein |

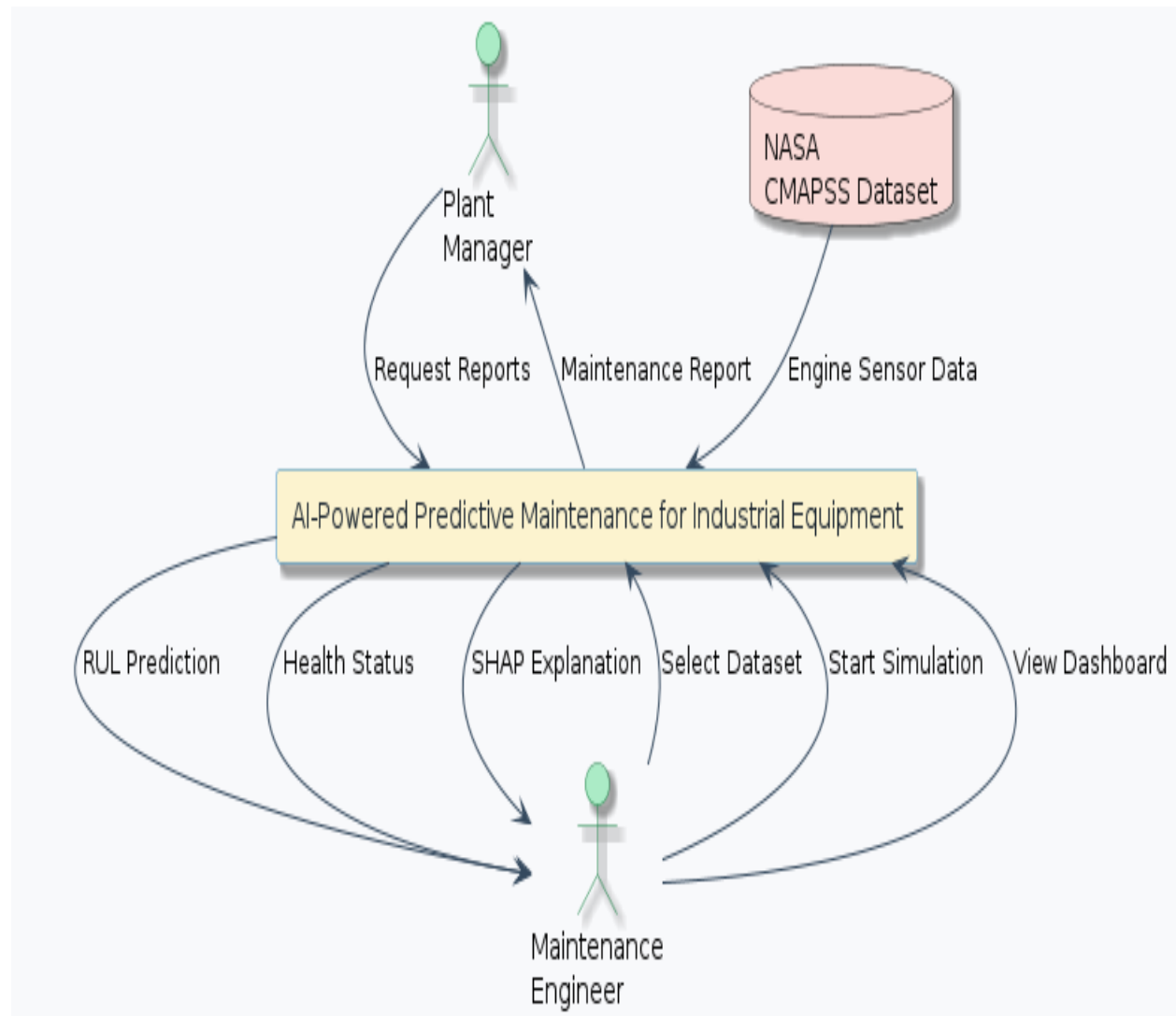
| <b>Task</b>                    | <b>Page No.</b> | <b>Evaluation</b> |
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| <b>Data Flow Diagram (DFD)</b> | <b>5</b>        |                   |
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| <b>State Diagram</b>           | <b>19</b>       |                   |
| <b>Class Diagram</b>           | <b>20</b>       |                   |



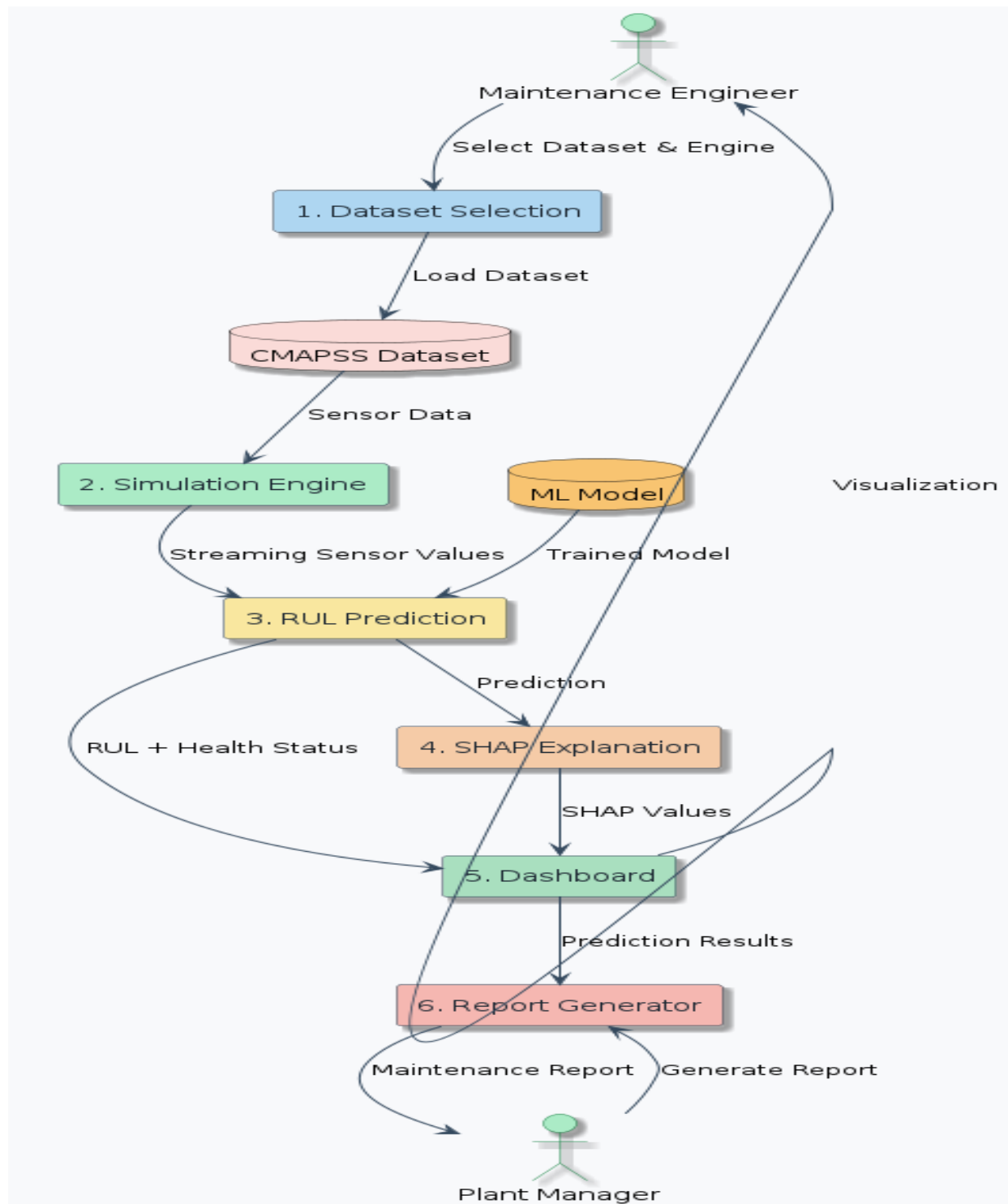
# **System Analysis & Design**

## Data Flow Diagram (DFD):

- DFD Level 0 (Context Diagram)

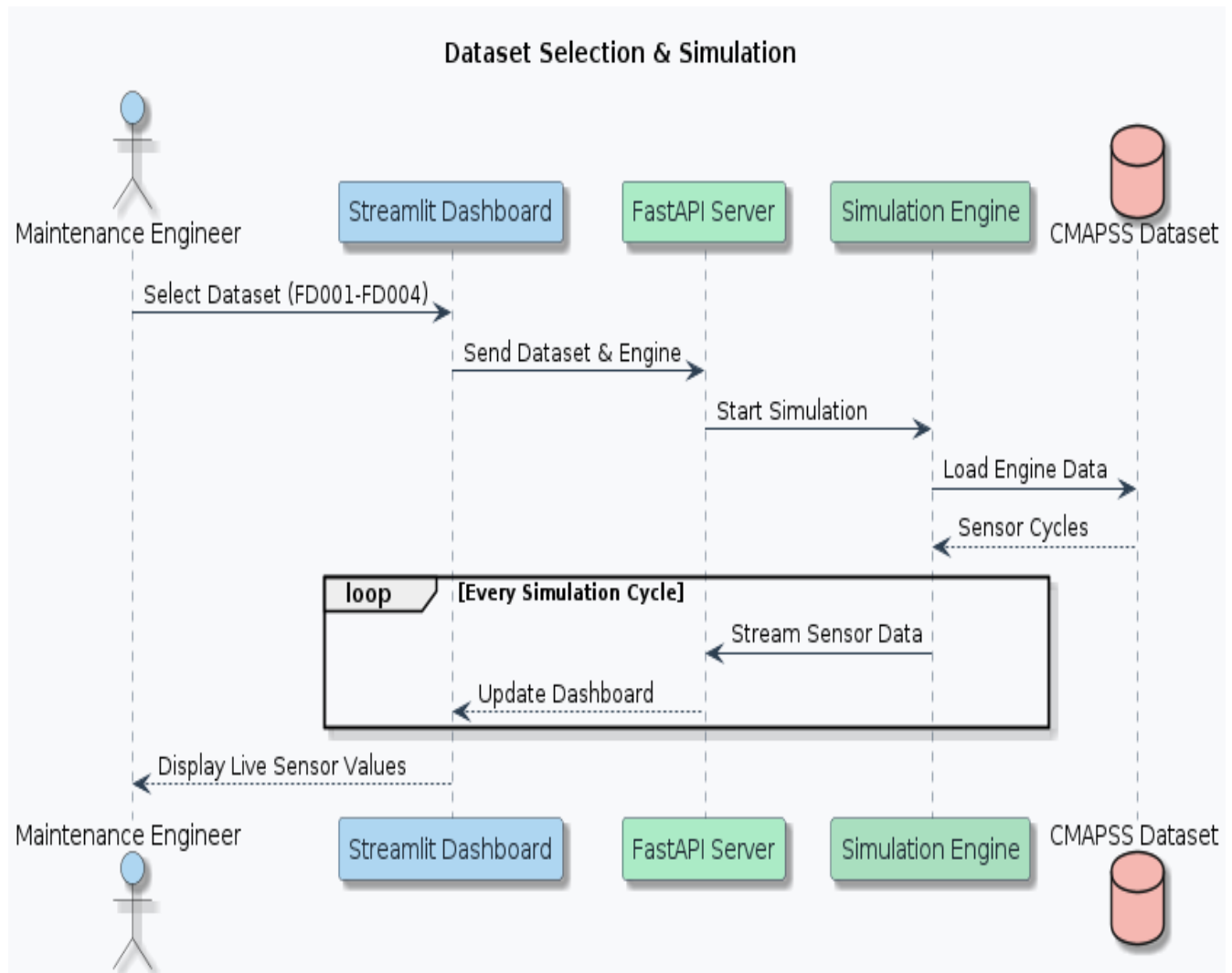


- DFD Level 1

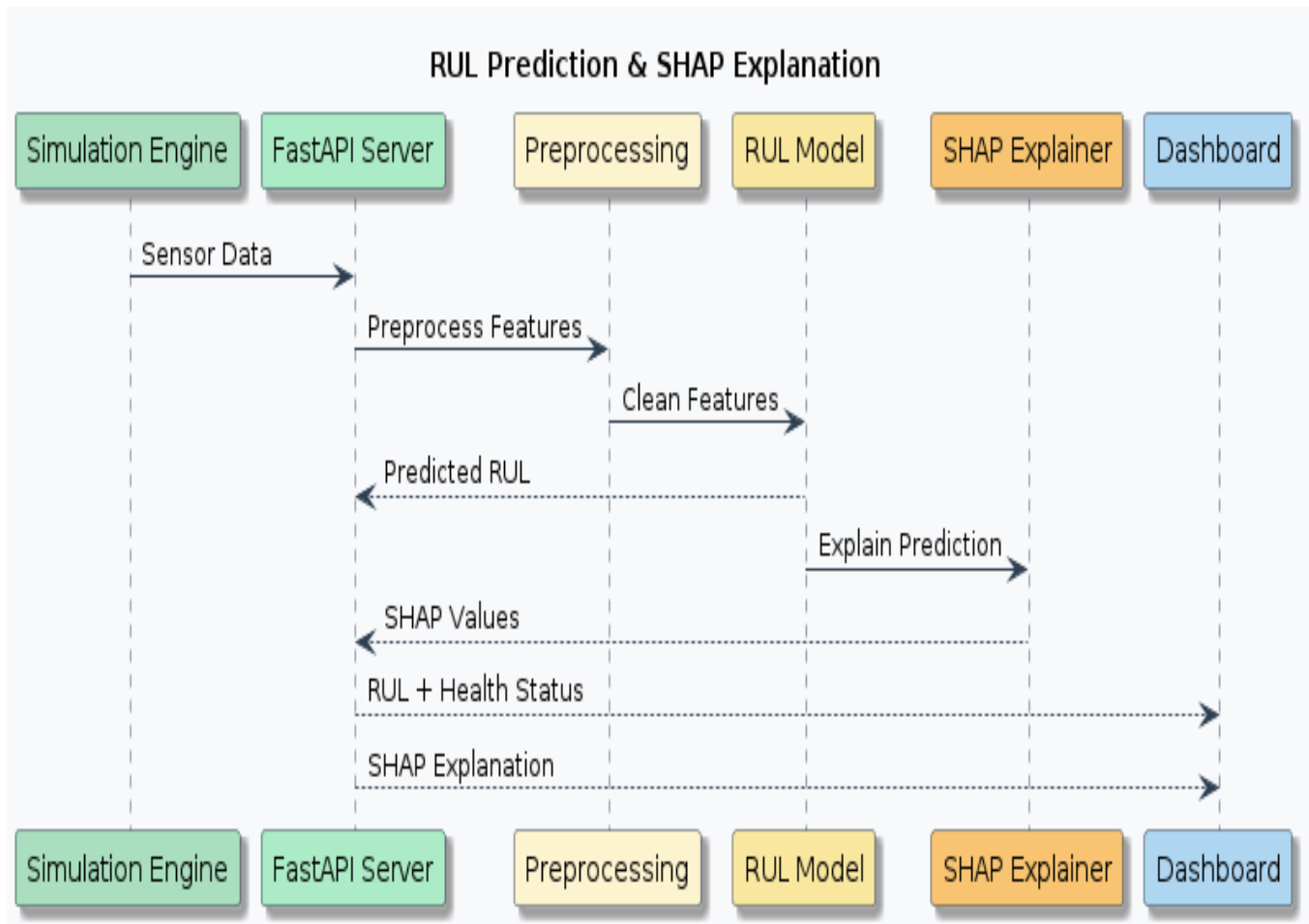


## Sequence Diagram:

- Sequence Diagram 1 — Dataset Selection & Simulation

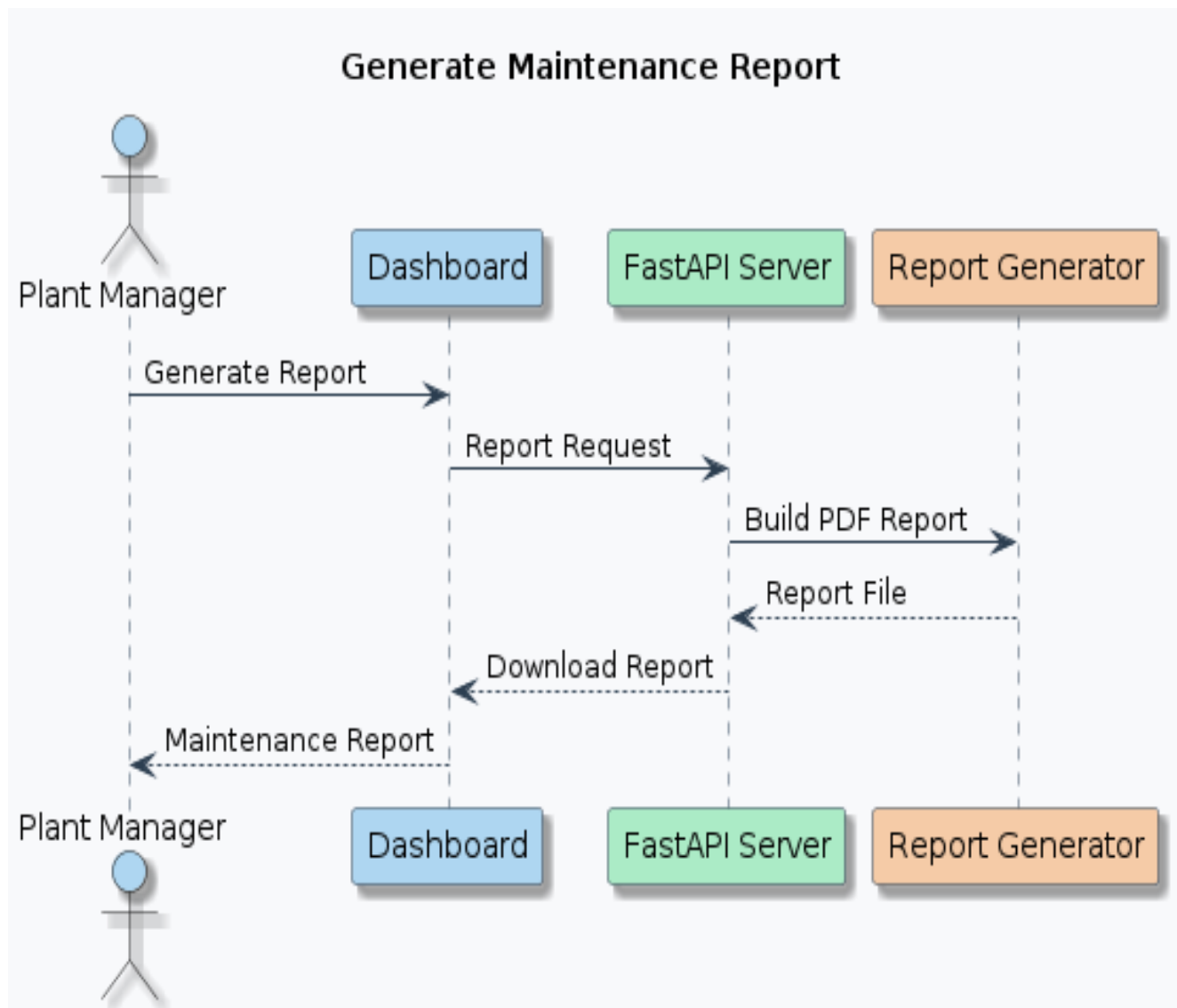


- Sequence Diagram 2 — RUL Prediction & SHAP Explanation

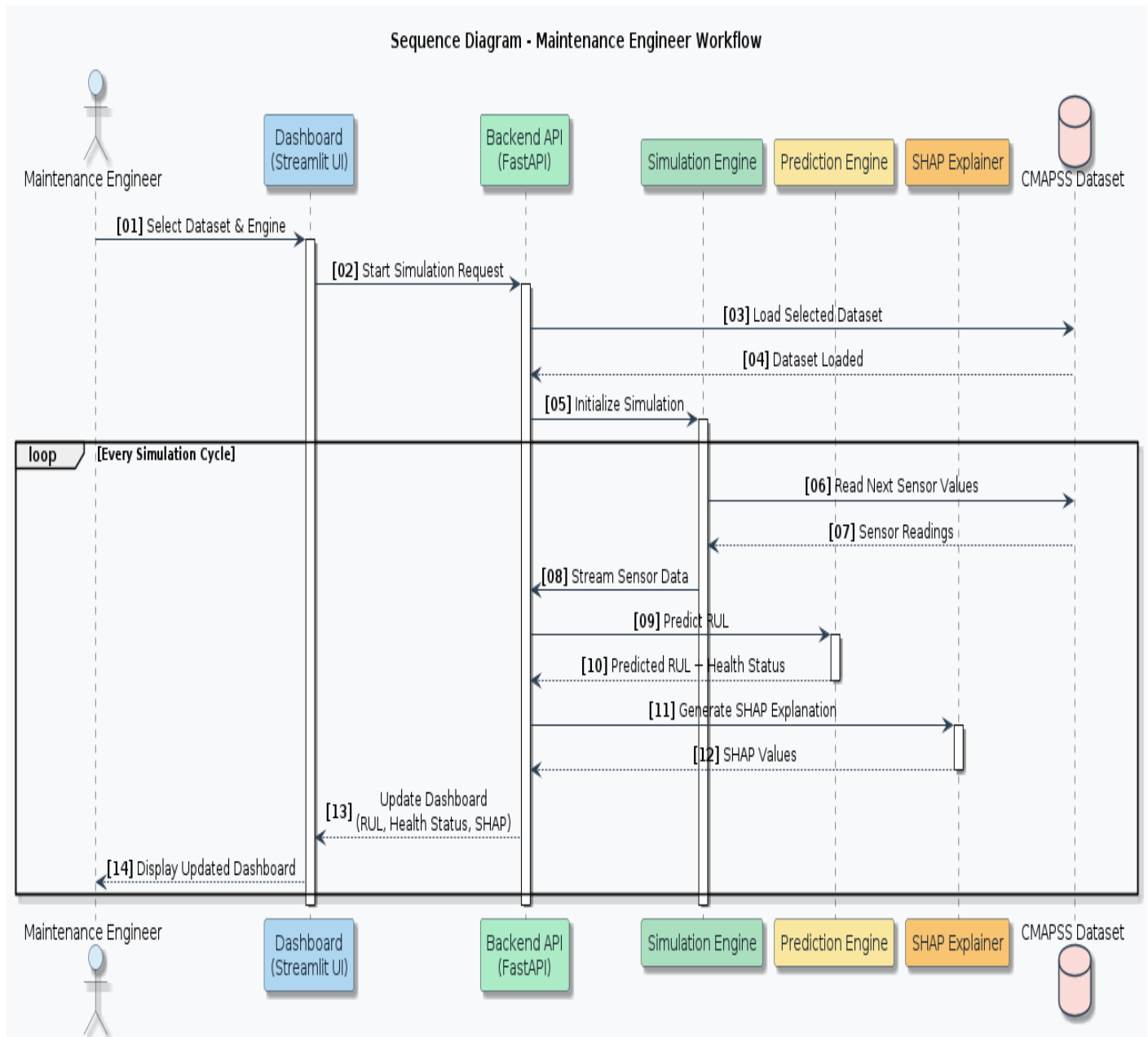




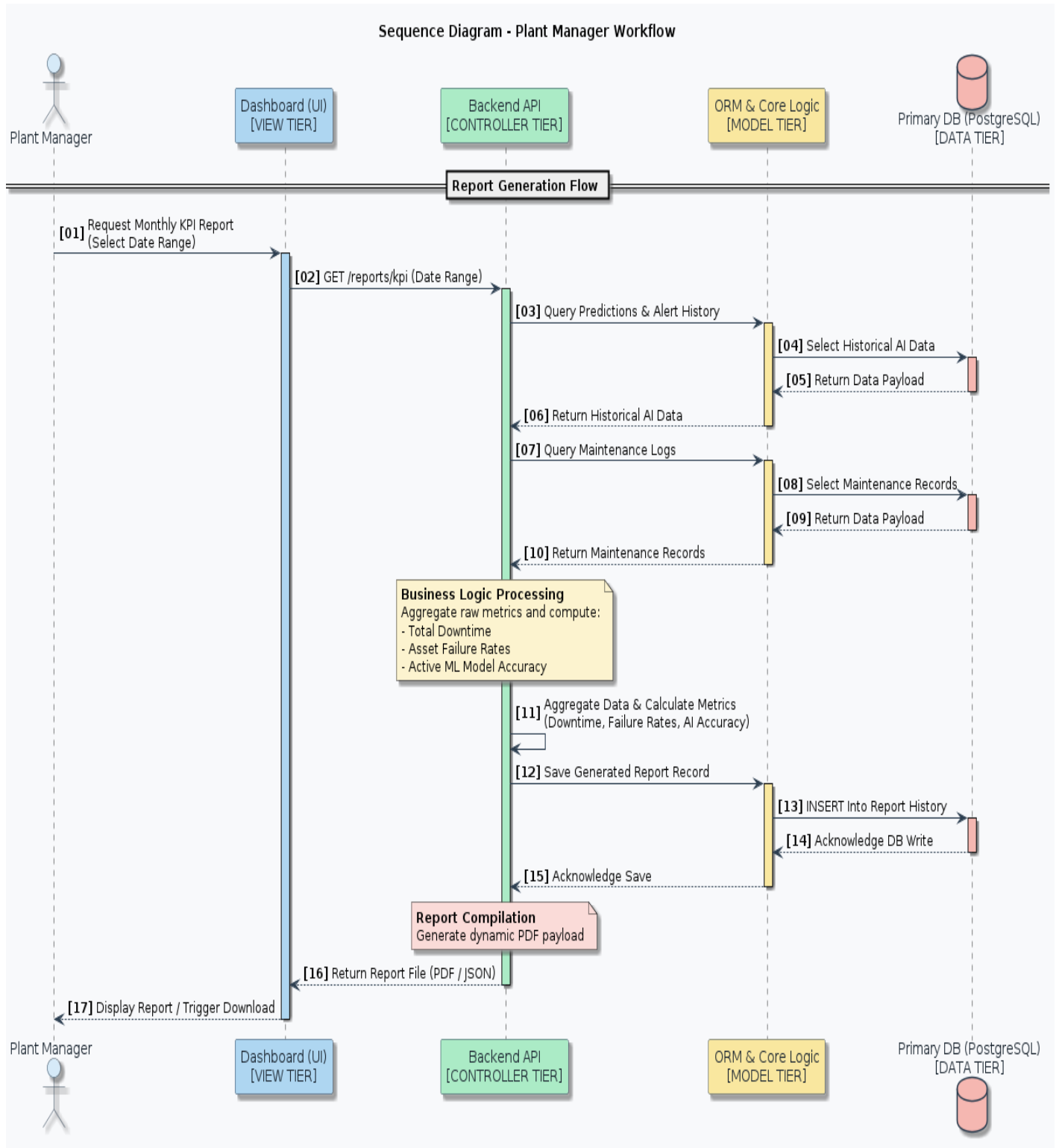
- **Sequence Diagram 3 — Generate Report**



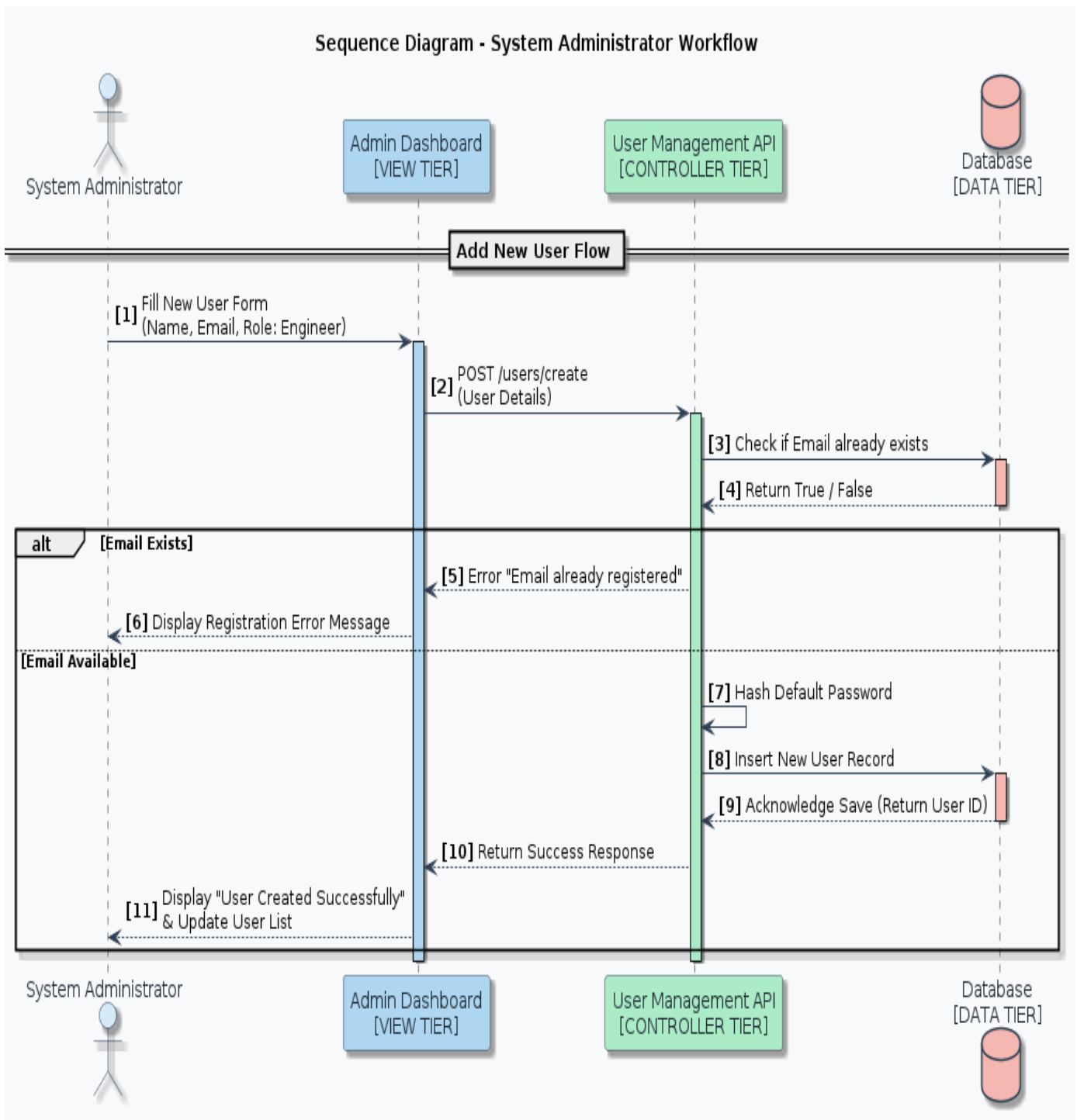
- Sequence Diagram 4 — Maintenance Engineer



## • Sequence Diagram 5 — Plant Manager

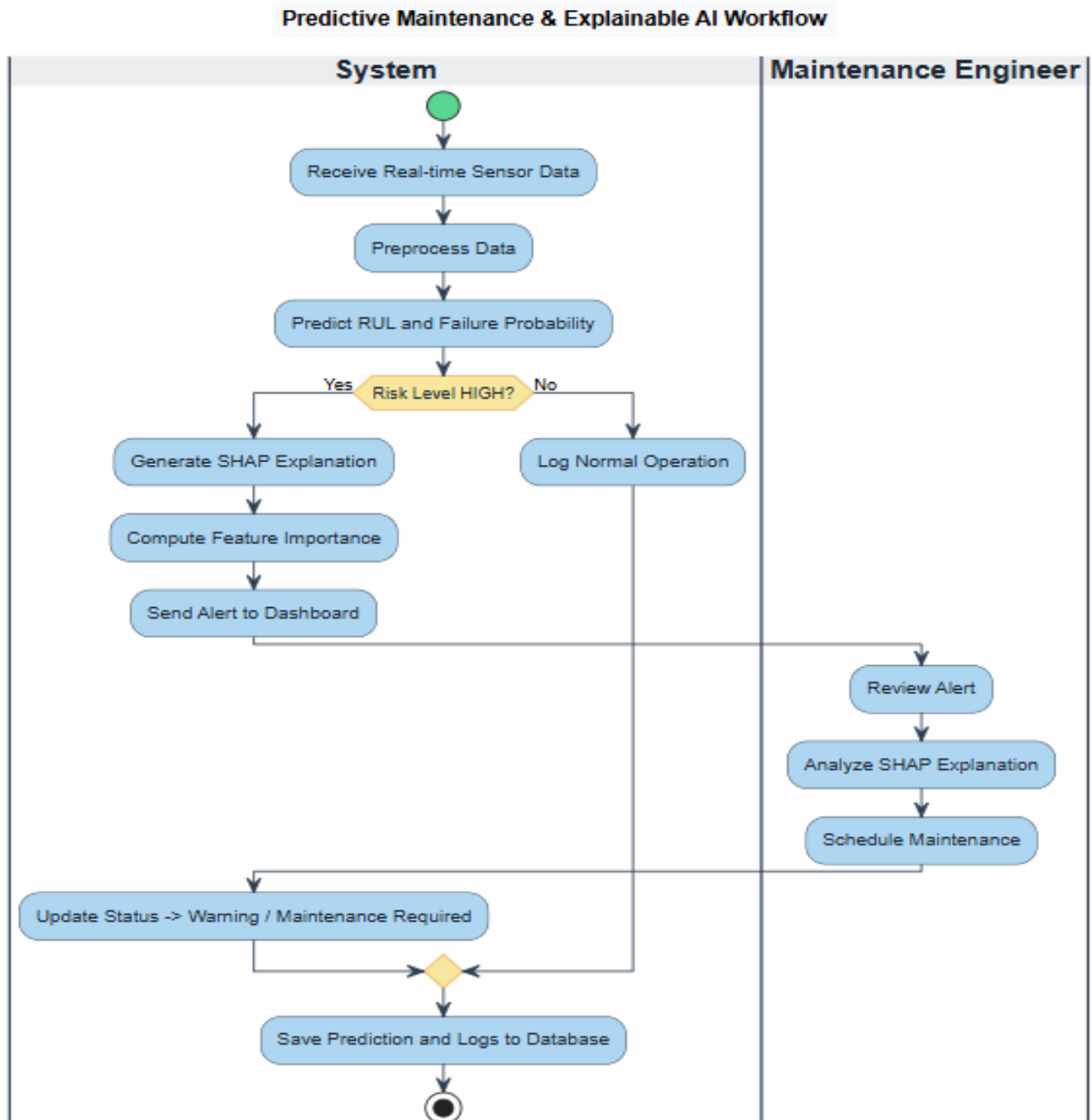


## • Sequence Diagram 6 — System Administrator



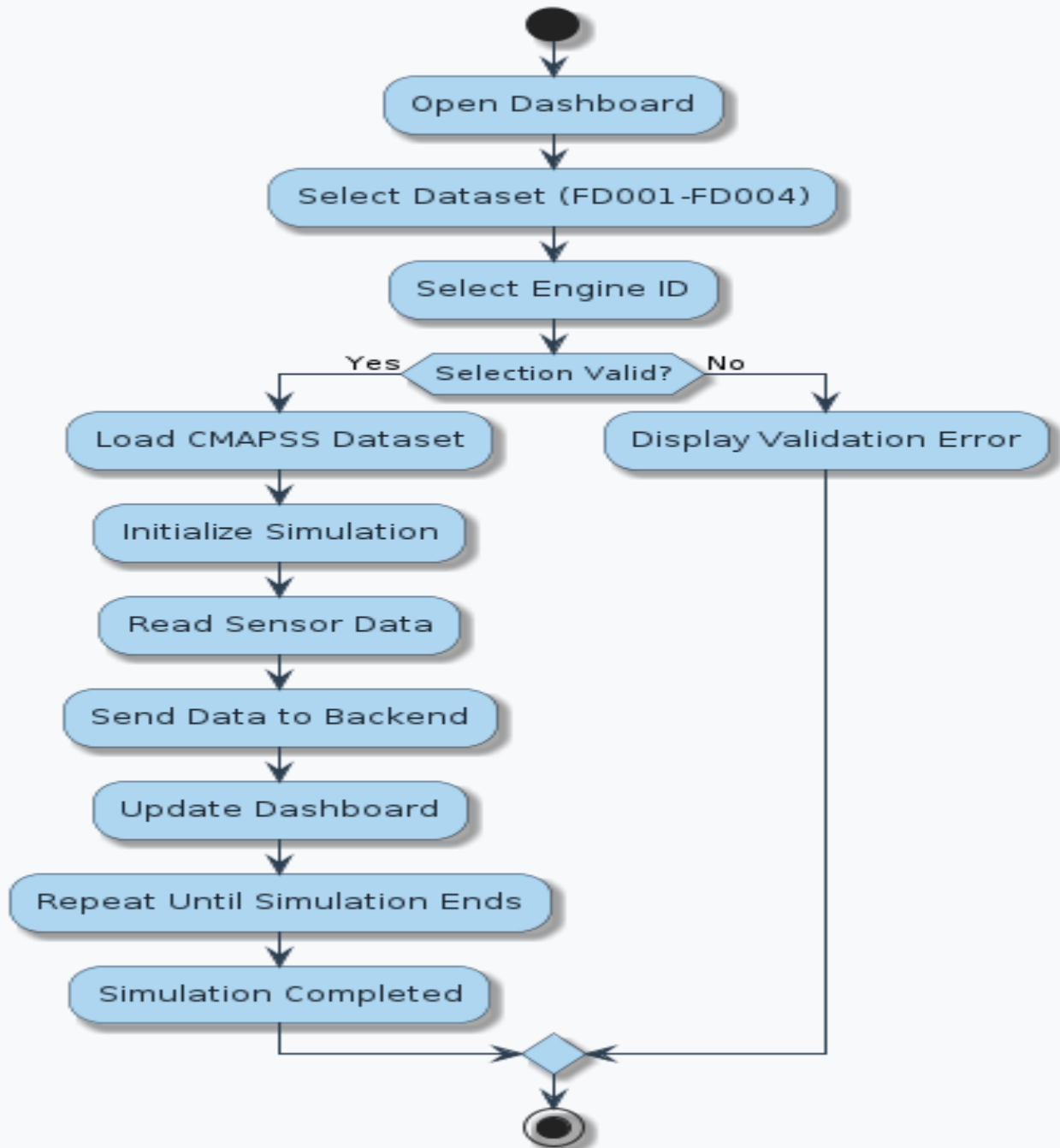
## Activity Diagram:

- Activity Diagram 1 — Predictive Maintenance

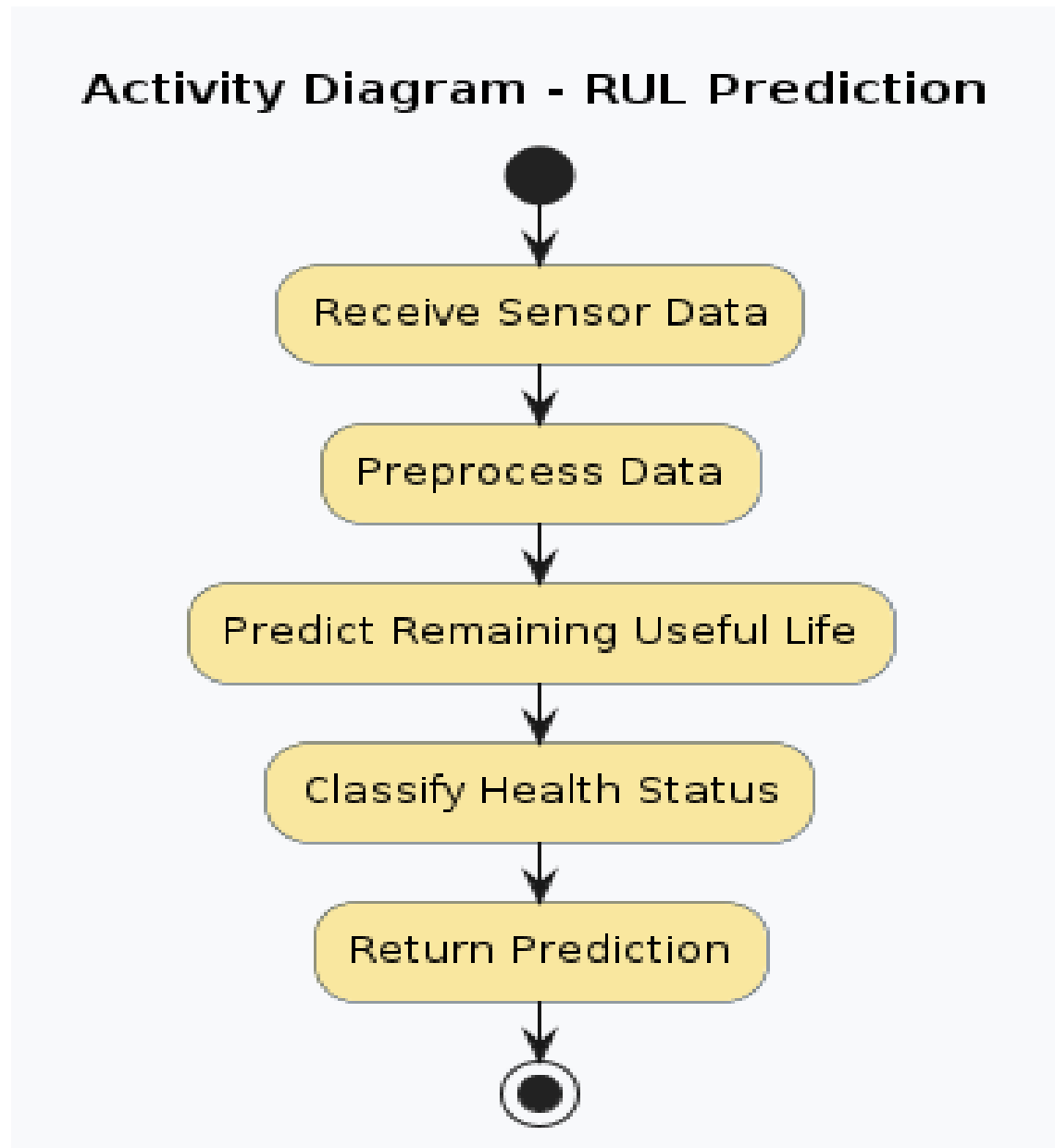


- Activity Diagram 2 — Dataset Selection & Simulation

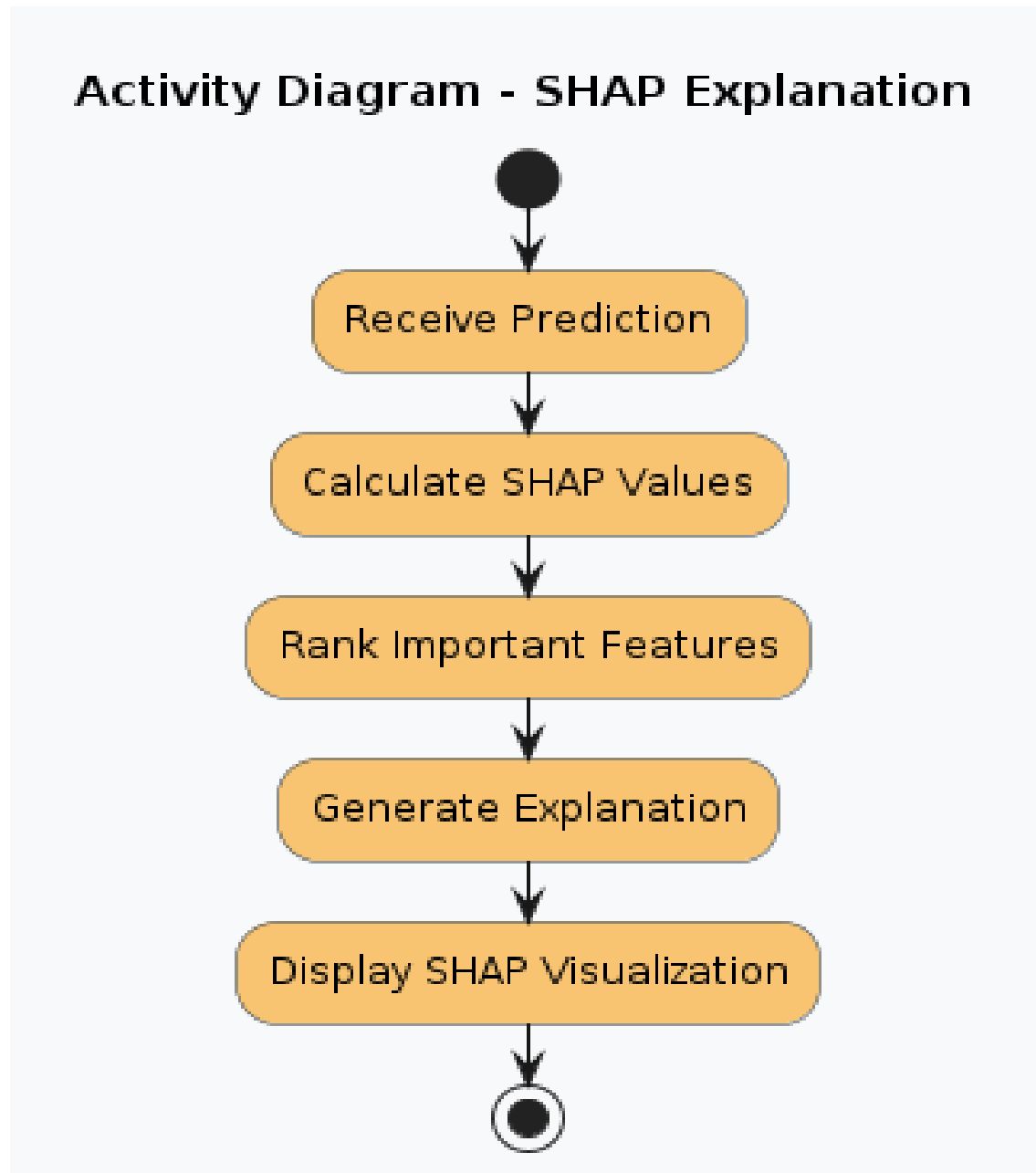
**Activity Diagram - Dataset Selection & Simulation**



- **Activity Diagram 3 — RUL Prediction**



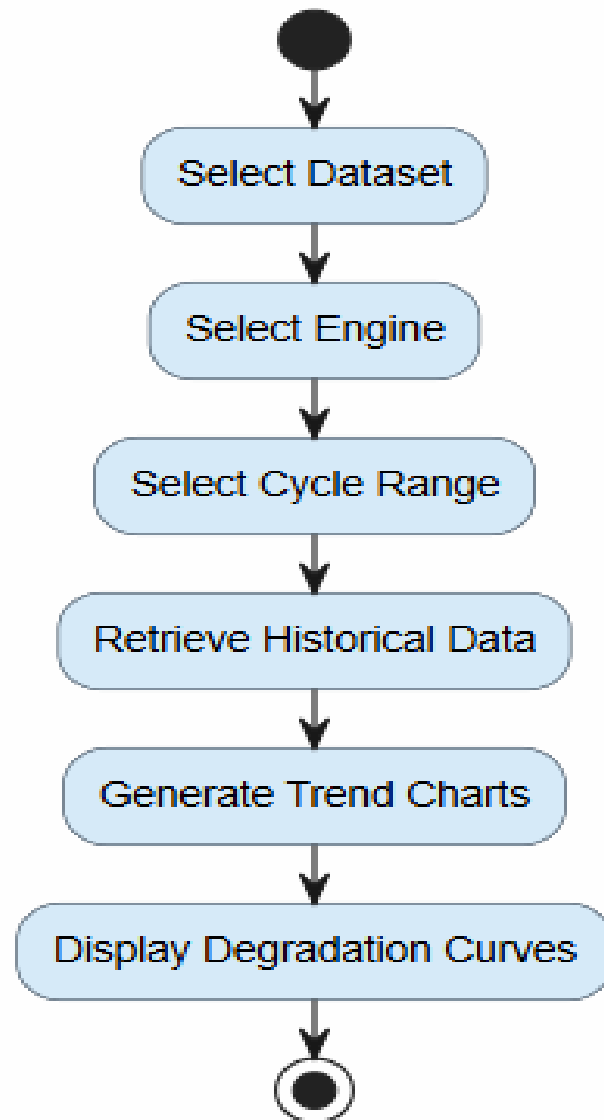
- Activity Diagram 4 — SHAP Explanation





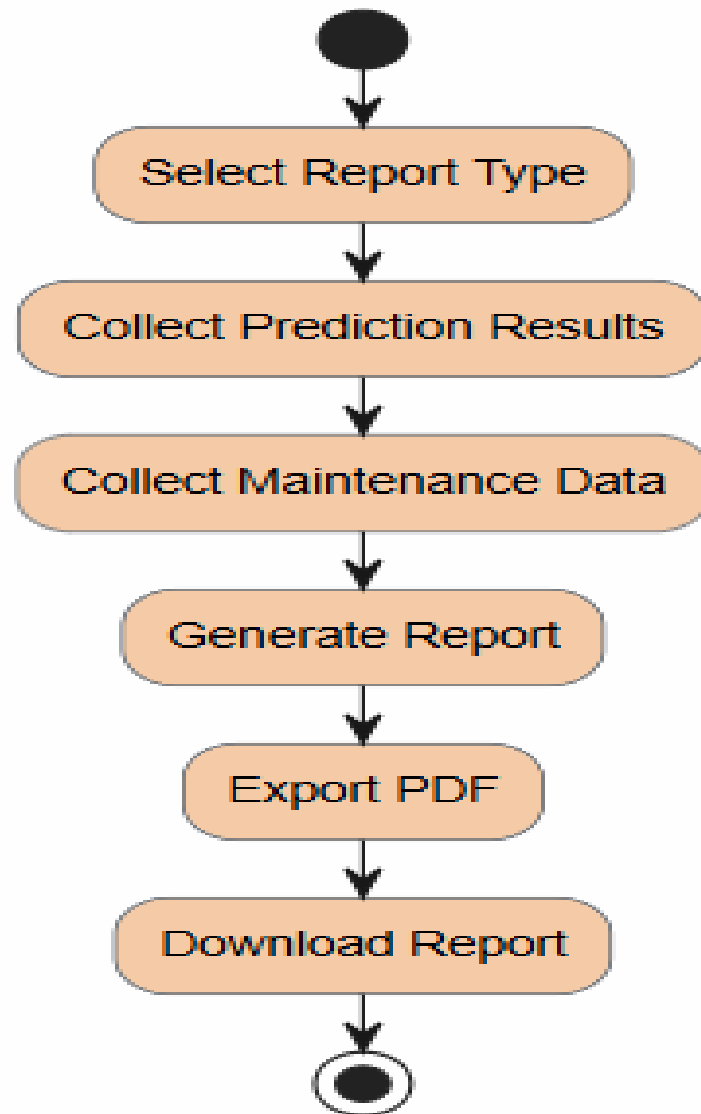
- Activity Diagram 5 — Historical Data Analysis

### Activity Diagram - Historical Data Analysis

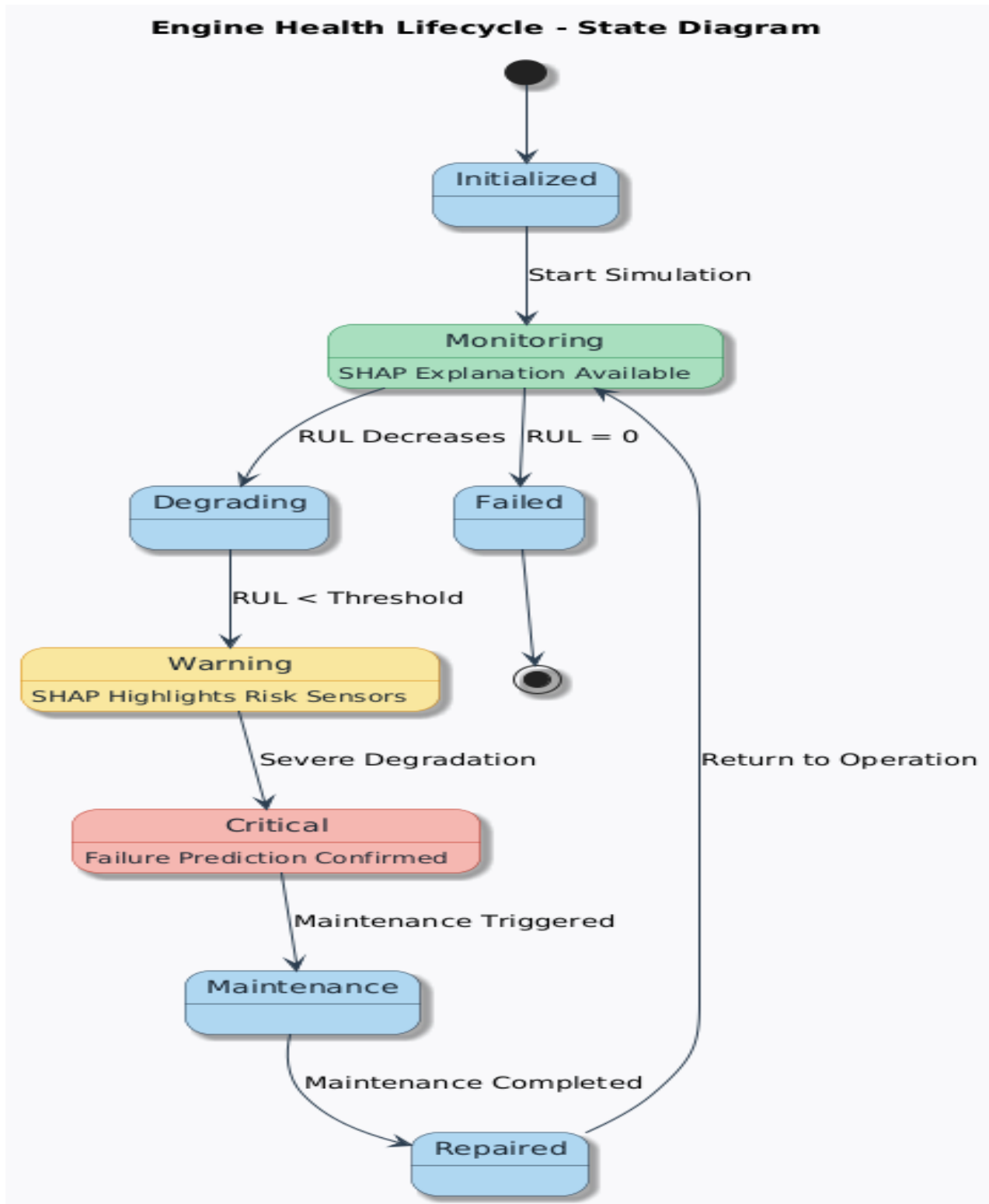


- Activity Diagram 6 — Report Generation

### Activity Diagram - Report Generation



## State Diagram:



## Class Diagram:

